

Similar Equations with different roots

$$(i) \quad x^2 + 5x + 6 = 0$$

$$1) \quad (1) \times (6) = +6$$

$$2) \quad \text{Multiply} = +6$$

$$\text{Addition} = +5$$

$$(3, 2)$$

$$3) \quad -3, -2$$

$$4) \quad -\frac{3}{1}, -\frac{2}{1} \Rightarrow (-3, -2)$$

Roots

$$(ii) \quad x^2 - 5x - 6 = 0$$

$$1) \quad (1) \times (-6) = -6$$

$$2) \quad \text{Multiply} = (-6)$$

$$\text{Subtract} = (-5)$$

$$(-6, +1)$$

$$3) \quad +6, -1$$

$$4) \quad +\frac{6}{1}, -\frac{1}{1} \Rightarrow (6, -1)$$

Roots

$$(iii) \quad x^2 + 5x - 6 = 0$$

$$1) \quad (1) \times (-6) = -6$$

$$2) \quad \text{Multiply} = -6$$

$$\text{Subtract} = +5$$

$$(6, -1)$$

$$3) \quad -6, +1$$

$$4) \quad -\frac{6}{1}, \frac{1}{1} \Rightarrow (-6, 1)$$

Roots

$$(iv) \quad x^2 - 5x + 6 = 0$$

$$1) \quad (1) \times (6) = +6$$

$$2) \quad \text{Multiply} = +6$$

$$\text{Addition} = -5$$

$$(-3, -2)$$

$$3) \quad +3, +2$$

$$4) \quad \frac{3}{1}, \frac{2}{1} \Rightarrow (3, 2)$$

Roots

$$(i) \quad y^2 - 14y + 48 = 0$$

$$1) \quad (1) \times (48) = +48$$

$$2) \quad \text{Multiply} = 48$$

$$\text{Addition} = -14$$

$$(-8, -6)$$

$$3) \quad +8, +6$$

$$4) \quad \frac{8}{1}, \frac{6}{1} \Rightarrow (8, 6)$$

Roots

$$(ii) \quad y^2 + 14y + 48 = 0$$

$$1) \quad (1) \times (48) = 48$$

$$2) \quad \text{Multiply} = 48$$

$$\text{Addition} = 14$$

$$(+8, +6)$$

$$3) \quad -8, -6$$

$$4) \quad -\frac{8}{1}, -\frac{6}{1} \Rightarrow (-8, -6)$$

Roots

$$(iii) \quad y^2 - 2y - 48 = 0$$

$$1) (1) \times (-48) = -48$$

$$2) \text{ Multiply} = -48$$

$$\text{Subtract} = -2$$

$$(-8, +6)$$

$$3) 8, -6$$

$$4) \frac{8}{1}, \frac{-6}{1} \Rightarrow (8, -6)$$

Roots

$$(iv) \quad y^2 + 2y - 48 = 0$$

$$1) (1) \times (-48) = -48$$

$$2) \text{ Multiply} = -48$$

$$\text{Subtract} = +2$$

$$(+8, -6)$$

$$3) -8, 6$$

$$4) \frac{-8}{1}, \frac{6}{1} \Rightarrow (-8, 6)$$

Roots

Understanding Signs :-

$$1) ax^2 - bx + c = 0$$

$$(+, +)$$

$$2) ax^2 + bx + c = 0$$

$$(-, -)$$

Both Roots will have same sign

$$3) ax^2 + bx - c = 0$$

$$(+, -)$$

$$4) ax^2 - bx - c = 0$$

$$(+, -)$$

Both roots will have opposite sign.

- When there is negative sign in constant term of a quadratic equation, both the signs of the roots will be opposite. like, $ax^2 + bx - c = 0$
 $ax^2 - bx - c = 0$

- When there is positive sign in constant term of the quadratic equation, both roots will have same sign.

→ when coefficient of x is negative = roots positive

→ when coefficient of x is positive = roots negative

Determining the sign of roots $\rightarrow$

Equation		Sign of Roots
$x^2 + 9x + 20 = 0$	constant sign (+)	(-, -)
$9x^2 - 45x + 36 = 0$	constant sign (+)	(+, +)
$1.8x^2 + 4x - 5 = 0$	constant sign (-)	(+, -)
$25x^2 - 72.5x + 25 = 0$	constant sign (+)	(+, +)
$20x^2 - 10x - 60 = 0$	constant sign (-)	(+, -)
$14x^2 + 35x - 21 = 0$	constant sign (-)	(+, -)
$25x^2 - 9x + 4 = 0$	constant sign (+)	(+, +)
$7x^2 + 2x - 9 = 0$	constant sign (-)	(+, -)
$0.25x^2 - 2x - 15 = 0$	constant sign (-)	(+, -)
$3x^2 + 13x + 10 = 0$	constant sign (+)	(-, -)
$4x^2 - 4x - 15 = 0$	constant sign (-)	(+, -)
$5x^2 + 5.5x - 6 = 0$	constant sign (-)	(+, -)
$6.5x^2 - 19x + 15 = 0$	constant sign (+)	(+, +)
$60x^2 - 31x + 24 = 0$	constant sign (+)	(+, +)
$11x^2 + 22x + 11 = 0$	constant sign (+)	(-, -)
$12x^2 - 3x - 6.75 = 0$	constant sign (-)	(+, -)
$13x^2 + 39x + 26 = 0$	constant sign (+)	(-, -)
$x^2 - 7x + 12 = 0$	constant sign (+)	(+, +)
$3x^2 + 8x - 9 = 0$	constant sign (-)	(+, -)
$2x^2 - 11x + 5 = 0$	constant sign (+)	(+, +)

$$(i) x^2 + x - 20 = 0$$

→ Multiply = $(1) \times (-20) = -20$

Subtract = $+1$

$(-4, 5)$

→ Constant sign $(-)$

Both roots will have opposite signs.

→ x coefficient sign $(+)$, so smallest root will be $(+)$

$(+4, -5)$ Roots.

$$(iii) x^2 - 9x + 20 = 0$$

Multiply → 20

Gap (Add) → 9

$(5, 4)$

Constant sign $(+)$

x coefficient sign $(-)$

$(+5, +4)$ Roots

$$(v) x^2 + 3x - 40 = 0$$

Multiply → 40

Gap → 3

→ Constant sign $(-)$

Both roots opposite sign

→ x coefficient $(+)$

$(+5, -8)$

$$(ii) x^2 - x - 20 = 0$$

→ Multiply = $(1) \times (-20) = -20$

Subtract = -1

$(4, -5)$

→ Constant sign $(-)$

Both roots will have opposite signs

→ x coefficient sign $(-)$

So, smallest root will be $(-)$

$(-4, +5)$ Roots.

$$(iv) x^2 + 9x + 20 = 0$$

Multiply → 20

Gap → 9 (Add)

$(4, 5)$

Constant sign $(+)$

Means same sign roots

x coefficient $(+)$

Means both roots $(-)$

$(-4, -5)$ Roots.

$$(vi) x^2 - 14x + 40 = 0$$

Multiply → 40

Gap (Add) → 14

→ Constant sign $(+)$

(Same sign roots)

x coefficient $(-)$

$(+10, +4)$ Roots.

When Coefficient of $x^2 \neq 1$

(i) $2x^2 - 4x - 48 = 0$

$2(x^2 - 2x - 48) = 0$

$x^2 - 2x - 48 = 0$

1) Product = 48

Difference = 2

2) constant sign (-)

Opposite sign roots

3) x coefficient sign (-)

Smallest root (-)

4) $(+8, -6)$ Roots

(iii) $9x^2 + 3x - 2 = 0$

1) Product = $9 \times 2 = 18$

Difference = 3

2) opposite signs

3) $(+3, -6)$

4) $\left(+\frac{3}{9}, -\frac{6}{9}\right)$ Divided by coefficient of x^2

5) $\left(+\frac{1}{3}, -\frac{2}{3}\right)$ Roots

(iv) $3x^2 - 17x + 10 = 0$

Product = 30

Sum = 17

Same sign roots

$\left(+\frac{15}{3}, +\frac{2}{3}\right) \rightarrow \left\{+5, +\frac{2}{3}\right\}$ Roots

(ii) $5x^2 - 16x + 3 = 0$

1) Product = $(5) \times (3) = 15$

Sum = 16

2) constant sign (+)

3) $+15, +1$

4) $+\frac{15}{5}, \frac{1}{5}$ Divide by coefficient of x^2

5) $\left(+3, +\frac{1}{5}\right)$ Roots

(iv) $25x^2 - 60x + 32 = 0$

1) Product = $25 \times 32 = 800$

Addition = 60

2) Same sign roots

3) $+40, +20$

4) $\left(+\frac{40}{25}, +\frac{20}{25}\right)$

5) $\left(+\frac{8}{5}, +\frac{4}{5}\right)$ Roots

(vi) $3x^2 - 26x + 16 = 0$

Product = $16 \times 3 = 48$

Sum = 26

$\left\{+\frac{24}{3}, +\frac{2}{3}\right\} \rightarrow \left\{+8, +\frac{2}{3}\right\}$ Roots

Learning Options → {Comparison of roots}

1) $x > y$ When both roots of x are greater than both roots of y .

for example, Roots of $x = 14, 12$ $\begin{cases} x_1 = 14 & x_2 = 12 \\ y_1 = 5 & y_2 = 8 \end{cases}$
 Roots of $y = 5, 8$

$$\begin{array}{lcl} \text{Now, } x_1 > y_1 & x_2 > y_1 & 14 > 5 \quad 12 > 5 \\ x_1 > y_2 & x_2 > y_2 & 14 > 8 \quad 12 > 8 \end{array} \Rightarrow$$

So, $x > y$

2) $x < y$ When both the roots of x are less than both roots of y .

for example, Roots of $x = 5, 8$ $\begin{cases} x_1 = 5 & x_2 = 8 \\ y_1 = 14 & y_2 = 10 \end{cases}$
 Roots of $y = 14, 10$

$$\begin{array}{lcl} \text{Now, } x_1 < y_1 & x_2 < y_1 & 5 < 14 \quad 8 < 14 \\ x_1 < y_2 & x_2 < y_2 & 5 < 10 \quad 8 < 10 \end{array} \Rightarrow$$

So, $x < y$

3) $x \geq y$ When the roots of x are either greater or equal to roots of y .

for example, Roots of $x = 13, 11$ $\begin{cases} x_1 = 13 & x_2 = 11 \\ y_1 = 5 & y_2 = 11 \end{cases}$
 Roots of $y = 5, 11$

$$\begin{array}{lcl} \text{Now, } x_1 > y_1 & x_2 > y_1 & 13 > 5 \quad 11 > 5 \\ x_1 > y_2 & x_2 = y_2 & 13 > 11 \quad 11 = 11 \end{array} \Rightarrow$$

So, $x \geq y$

4) $x \leq y$

When the roots of x are lesser or equal to the roots of y .

for example,

$$\begin{array}{l} \text{Roots of } x = 5, 11 \\ \text{Roots of } y = 13, 11 \end{array} \quad \left\{ \begin{array}{ll} x_1 = 5 & x_2 = 11 \\ y_1 = 13 & y_2 = 11 \end{array} \right\}$$

$$\begin{array}{ll} \text{Now, } x_1 < y_1 & x_2 < y_1 \\ x_1 < y_2 & x_2 = y_2 \end{array} \Rightarrow \begin{array}{ll} 5 < 13 & 11 < 13 \\ 5 < 11 & 11 = 11 \end{array}$$

$$\boxed{\text{So, } x \leq y}$$

5) $x = y$

When both the roots of x are exactly equal to both the roots of y .

$$\begin{array}{l} \text{for example, Roots of } x = 5, 5 \\ \text{Roots of } y = 5, 5 \end{array}$$

$$\text{Now, } 5 = 5 = 5 = 5$$

$$\boxed{\text{So, } x = y}$$

6) Relation Not Possible Situation

When while comparing the roots of x and y , we find both the situations of $x < y$ and $x > y$, then the relation between roots is not possible.

for example,

$$\begin{array}{l} \text{Roots of } x = 5, 10 \\ \text{Roots of } y = 7, 12 \end{array} \quad \left\{ \begin{array}{ll} x_1 = 5 & x_2 = 10 \\ y_1 = 7 & y_2 = 12 \end{array} \right\}$$

$$\begin{array}{ll} \text{Now, } x_1 < y_1 & x_2 > y_1 \\ x_1 < y_2 & x_2 < y_2 \end{array} \Rightarrow \begin{array}{ll} 5 < 7 & 10 > 7 \quad (x > y) \\ 5 < 12 & 10 < 12 \quad (x < y) \end{array}$$

So, Relation cannot be possible.

Practice Questions →

$$(I) x^2 - 9x + 18 = 0$$

$$\text{Product} = 18$$

$$\text{Sum} = -9$$

→ Same sign roots
(+6, +3)

Comparison of roots →

$$\text{Roots of } x = 6, 3$$

$$\text{Roots of } y = 12, 10$$

$$\left\{ \begin{array}{ll} 6 < 12 & 3 < 12 \\ 6 < 10 & 3 < 10 \end{array} \right\}$$

$$\boxed{\text{So, } x < y}$$

$$(I) x^2 - 45x + 500 = 0$$

$$\text{Product} = 500$$

$$\text{Sum} = 45$$

→ Same sign roots
(+25, +20)

Comparison of roots →

$$\text{Roots of } x = 25, 20$$

$$\text{Roots of } y = 18, -6$$

$$\left\{ \begin{array}{ll} 25 > 18 & 20 > 18 \\ 25 > -6 & 20 > -6 \end{array} \right\}$$

$$\boxed{\text{So, } x > y}$$

$$(I) x^2 - x - 380 = 0$$

$$\text{Product} = 380$$

$$\text{Difference} = 1$$

→ Roots with opposite signs
(-19, 20)

Comparison :-

$$\left\{ \begin{array}{ll} \text{Roots of } x = -19, 20 & \begin{array}{ll} -19 < 22 & 20 < 22 \\ -19 < 20 & 20 = 20 \end{array} \end{array} \right.$$

$$\boxed{\text{So, } x \leq y}$$

$$(II) y^2 - 22y + 120 = 0$$

$$\text{Product} = 120$$

$$\text{Sum} = -22$$

→ Same sign roots
(+12, +10)

$$(II) y^2 - 12y - 108 = 0$$

$$\text{Product} = 108$$

$$\text{Difference} = 12$$

→ Roots with opposite signs
(+18, -6)

$$(II) y^2 - 42y + 440 = 0$$

$$\text{Product} = 440$$

$$\text{Sum} = 42$$

→ Same sign roots
(22, 20)

$$(I) x^2 - 13x + 40 = 0$$

$$\text{Product} = 40$$

$$\text{Sum} = 13$$

→ Same sign roots
(8, 5)

$$(II) 2y^2 + 3y - 9 = 0$$

$$\text{Product} = (2)(9) = 18$$

$$\text{Difference} = 3$$

→ Roots with different signs.

$$\left(-\frac{6}{2}, +\frac{3}{2}\right)$$

$$\Rightarrow (-3, +1.5)$$

Comparison of Roots →

$$\begin{array}{l} \text{Roots of } x = 8, 5 \\ \text{Roots of } y = -3, 1.5 \end{array} \left\{ \begin{array}{ll} 8 > -3 & 5 > -3 \\ 8 > 1.5 & 5 > 1.5 \end{array} \right\}$$

$$\boxed{\text{So, } x > y}$$

Option 2 ↘

$$\begin{array}{l} \text{Roots of } x \\ (8, 5) \end{array}$$

$$\begin{array}{l} \text{Roots of } y \\ \left(-\frac{6}{2}, +\frac{3}{2}\right) \end{array}$$

Either divide roots of y by 2
or Multiply roots of x by 2

Now,

$$\text{Roots of } x = \left(\frac{8}{x_2}, \frac{5}{x_2}\right) \Rightarrow (16, 10)$$

$$\text{Roots of } y = (-6, +3)$$

Comparison :-

$$\left\{ \begin{array}{ll} 16 > -6 & 10 > -6 \\ 16 > 3 & 10 > 3 \end{array} \right\}$$

$$\boxed{x > y} \checkmark$$

$$(I) 3x^2 - 26x + 16 = 0$$

$$\text{Product} = (3)(16) = 48$$

$$\text{Sum} = 26$$

→ Same sign roots

$$\left(\frac{+24}{3}, \frac{+2}{3} \right)$$

$$(24, 2)$$

$$(II) y^2 - 23y + 120 = 0$$

$$\text{Product} = 120$$

$$\text{Sum} = 23$$

→ Same sign roots

$$(15, 8) \times 3$$

$$(45, 24)$$

Comparison: $\left. \begin{array}{l} \text{Roots of } x = 24, 2 \\ \text{Roots of } y = 45, 24 \end{array} \right\} \begin{array}{ll} 24 < 45 & 2 < 45 \\ 24 = 24 & 2 < 24 \end{array}$

$$\boxed{\text{So, } x \leq y}$$

$$(I) x^2 - 16x + 55 = 0$$

$$\text{Product} = 55$$

$$\text{Sum} = 16$$

$$(+11, +5) \times 5$$

$$(55, 25)$$

$$(II) 5y^2 - 33y + 40 = 0$$

$$\text{Product} = (40)(5) = 200$$

$$\text{Sum} = 33$$

$$\left(\frac{+25}{5}, \frac{+8}{5} \right)$$

$$(25, 8)$$

Comparison: $\left. \begin{array}{l} \text{Roots of } x = 55, 25 \\ \text{Roots of } y = 25, 8 \end{array} \right\} \begin{array}{ll} 55 > 25 & 25 = 25 \\ 55 > 8 & 25 > 8 \end{array}$

$$\boxed{\text{So, } x \geq y}$$

$$(I) \quad 9x^2 + 81x + 72 = 0$$

$$9(x^2 + 9x + 8) = 0$$

$$x^2 + 9x + 8 = 0$$

$$\text{Product} = 8$$

$$\text{Sum} = 9$$

→ Sign negative

$$(-8, -1)$$

$$(II) \quad y^2 + 6y + 8 = 0$$

$$\text{Product} = 8$$

$$\text{Sum} = 6$$

→ Sign negative

$$(-4, -2)$$

Comparison of Roots :-

$$\begin{array}{l} \text{Roots of } x = (-8, -1) \\ \text{Roots of } y = (-4, -2) \end{array} \left\{ \begin{array}{ll} -8 < -4 & -1 > -4 \\ -8 < -2 & -1 > -2 \end{array} \right\}$$

No, Relation can be determined

$$(I) \quad 7x^2 - 29x + 30 = 0$$

$$\text{Product} = (7)(30) = 210$$

$$\text{Sum} = 29$$

→ Sign positive

$$\left(\frac{+14}{7}, \frac{+15}{7} \right)$$

$$(14, 15)$$

$$(II) \quad y^2 + 4y - 12 = 0$$

$$\text{Product} = 12$$

S,

→ opposite sign

$$(-6, +2) \times 7$$

$$(-42, 14)$$

Comparison of Roots :-

$$\begin{array}{l} \text{Roots of } x = (14, 15) \\ \text{Roots of } y = (-42, 14) \end{array} \left\{ \begin{array}{ll} 14 > -42 & 15 > -42 \\ 14 = 14 & 15 > 14 \end{array} \right\}$$

So, $x \geq y$

$$(I) x^2 + 46x + 45 = 0$$

$$\text{Product} = 45$$

$$\text{Sum} = 46$$

→ Sign negative

$$(-45, -1)$$

Comparison of roots

$$\text{Roots of } x = (-45, -1)$$

$$\text{Roots of } y = (4, -0.04)$$

$$(II) 25y^2 - 99y - 4 = 0$$

$$\text{Product} = (25)(4) = 100$$

$$\text{Difference} = 99$$

→ opposite sign

$$\left(\frac{+100}{25}, -\frac{1}{25}\right) \Rightarrow (+4, -0.04)$$

$$\left. \begin{array}{l} \text{Roots of } x = (-45, -1) \\ \text{Roots of } y = (4, -0.04) \end{array} \right\} \begin{array}{ll} -45 < 4 & -1 < 4 \\ -45 < -0.04 & -1 < -0.04 \end{array}$$

$$\boxed{\text{So, } x < y}$$

$$(I) 3x^2 - 26x - 9 = 0$$

$$\text{Product} = 27$$

$$\text{Difference} = 26$$

$$\left(\frac{+27}{3}, -\frac{1}{3}\right)$$

$$(II) y^2 + 8y + 12 = 0$$

$$\text{Product} = 12$$

$$\text{Sum} = 8$$

$$\begin{array}{l} (-6, -2) \times 3 \\ (-18, -6) \end{array}$$

Comparison:

$$\left. \begin{array}{l} \text{Roots of } x = 27, -1 \\ \text{Roots of } y = -18, -6 \end{array} \right\} \begin{array}{ll} 27 > -18 & -1 > -18 \\ 27 > -6 & -1 > -6 \end{array}$$

$$\boxed{\text{So, } x > y}$$

$$(I) 8x^2 + 29x + 15 = 0$$

$$\text{Product} = 120$$

$$\text{Sum} = 29$$

$$\left(-\frac{24}{8}, -\frac{5}{8}\right)$$

$$(II) y^2 + 13y + 36 = 0$$

$$\text{Product} = 36$$

$$\text{Sum} = 13$$

$$\begin{array}{l} (-4, -9) \times 8 \\ (-32, -72) \end{array}$$

Comparison:

$$\left. \begin{array}{l} \text{Roots of } x = -24, -5 \\ \text{Roots of } y = -32, -72 \end{array} \right\} \begin{array}{ll} -24 > -32 & -5 > -72 \\ -24 > -72 & -5 > -72 \end{array}$$

$$\boxed{\text{So, } x > y}$$